Project: Worked examples
Title: Column buckling (Perry–Robertson)
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Calculates the compressive strength and resistance of a pin-ended strut by the Perry–Robertson formula, and its utilisation against an applied axial load. The Robertson constant comes from the strut curve.

WHAT IT CALCULATES	RELATION
Radius of gyration and slenderness	$r = \sqrt{(I/A)}$; $\lambda = L_E/r$
Euler buckling stress and load	$p_E = \pi^2 E/\lambda^2$; $F_E = p_E \cdot A$
Limiting slenderness, below which the strut is not reduced for buckling	$\lambda_0 = 0.2 \cdot \sqrt{(\pi^2 E/p_y)}$
Perry factor (Robertson constant a = 2.0, 3.5, 5.5, 8.0 for curves a–d)	$\eta = \max(0.001 \cdot a \cdot (\lambda - \lambda_0), 0)$
Compressive strength by the Perry–Robertson formula	$\varphi = (p_y + (\eta + 1) \cdot p_E)/2$; $p_c = p_E \cdot p_y/(\varphi + \sqrt{(\varphi^2 - p_E \cdot p_y)})$
Compression resistance and utilisation	$F_{c,Rd} = p_c \cdot A$; $u = F/F_{c,Rd}$

Basis: Compressive strength (Perry strut formula)[1](Annex C); Maximum slenderness[1](cl. 4.7.3.2); Allocation of strut curve[1](Table 23).

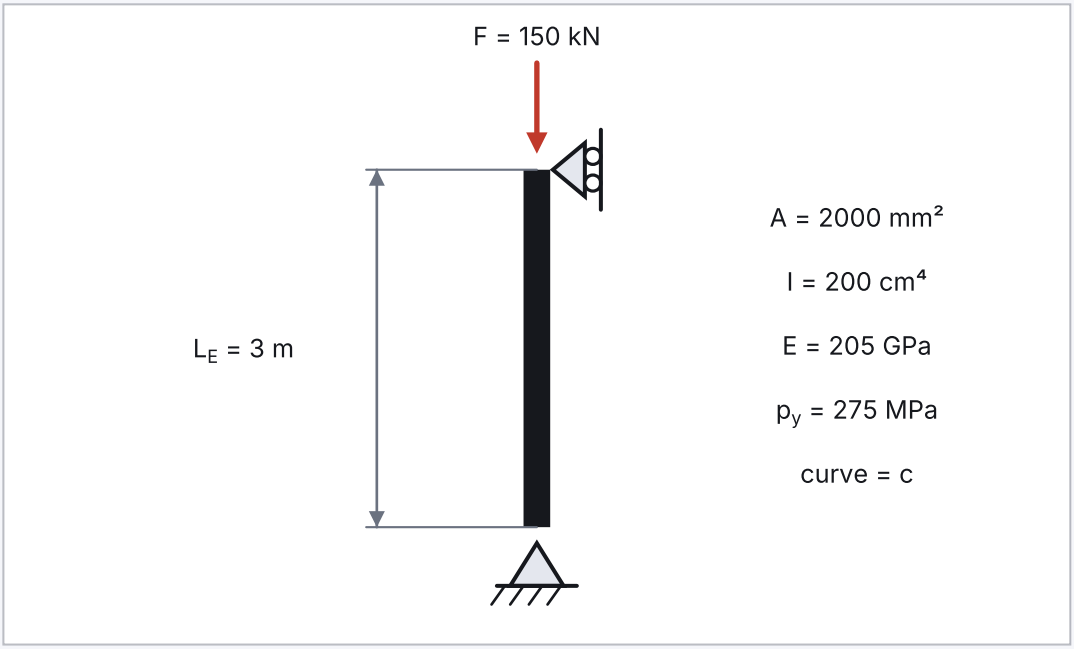


Figure 1 — A pin-ended strut under axial compression, effective length L_E. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
curve	Strut curve (selects the Robertson constant)	c		spec: BS 5950-1 Table 23: welded box, curve c	any

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
L_E	Effective length	3	m	drawing: DRG-ST-410 rev B	0.1 … 50 m
A	Cross-section area	2000	mm²	spec: SPEC-ST-02 section schedule	10 … 1000000 mm²
I	Second moment of area about the weaker axis	200	cm⁴	spec: SPEC-ST-02 section schedule	0.01 … 10000000 cm⁴
E	Young's modulus	205	GPa	spec: BS 5950-1 3.1.3: E = 205 GPa	50 … 450 GPa
p_y	Design strength (yield)	275	MPa	spec: SPEC-ST-02: S275, t ≤ 16 mm	100 … 1000 MPa
F	Applied axial load	150	kN	assumption: Factored axial load from the frame analysis	0 … 100000 kN

3. Assumptions

A1. The strut is pin-ended and prismatic, under axial load only. (Effective length carries the end restraint.)
A2. Buckling is about the axis of the given second moment. (The user gives the weaker axis.)
A3. Initial imperfection represented by the Perry factor $\eta = 0.001 \cdot a \cdot (\lambda - \lambda_0)$, with a from the strut curve. (Robertson's fit to strut tests.)
A4. F (Applied axial load) is taken as 150 kN. (Factored axial load from the frame analysis)

4. Working

[8] $a_R = 5.5$ [1](Annex C)
= 5.5
Robertson constant for strut curve c
[9] r = sqrt(I / A)
= sqrt(200 cm⁴ / 2000 mm²)
= 31.623 mm
Radius of gyration
[10] $\lambda = L_E$ / r
= 3 m / 31.623 mm
= 94.868
Slenderness
[11] p_E = pi^2 * E / λ^2 [1](Annex C)
= π^2 × 205 GPa / 94.868^2
= 224.81 MPa
Euler critical stress
[12] $F_E = p_E$ * A
= 224.81 MPa × 2000 mm²
= 449.62 kN
Euler critical load
[13] λ_0 = 0.2 * sqrt(pi^2 * E / p_y) [1](Annex C)
= 0.2 × sqrt(π^2 × 205 GPa / 275 MPa)
= 17.155
Limiting slenderness
[14] η = max(0.001 * a_R * (λ - λ_0), 0) [1](Annex C)
= max(0.001 × 5.5 × (94.868 - 17.155), 0)
= 0.42742
Perry factor (initial imperfection)

[15] $\varphi = (p_y + (\eta + 1) * p_E) / 2$
$= (275\ \text{MPa} + (0.42742 + 1) \times 224.81\ \text{MPa}) / 2$
$= 297.95\ \text{MPa}$
Perry–Robertson φ

[16] $p_c = p_E * p_y / (\varphi + \sqrt{\varphi^2 - p_E * p_y})$ [1](Annex C)
$= 224.81\ \text{MPa} \times 275\ \text{MPa} / (297.95\ \text{MPa} + \sqrt{(297.95\ \text{MPa})^2 - 224.81\ \text{MPa} \times 275\ \text{MPa}})$
$= 133.78\ \text{MPa}$
Compressive strength

[17] $F_{c,Rd} = p_c * A$
$= 133.78\ \text{MPa} \times 2000\ \text{mm}^2$
$= 267.56\ \text{kN}$
Compression resistance

[18] $u = F / F_{c,Rd}$
$= 150\ \text{kN} / 267.56\ \text{kN}$
$= 0.56062$
Utilisation

5. Results

SYMBOL	VALUE	DESCRIPTION
$F_{c,Rd}$	267.56 kN	Compression resistance
F_E	449.62 kN	Euler critical load
p_c	133.8 MPa	Compressive strength
λ	94.868	Slenderness
u	0.56	Utilisation (governing)
r	31.62 mm	Radius of gyration
p_E	224.8 MPa	Euler stress
λ_0	17.155	Limiting slenderness
η	0.42742	Perry factor

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Axial load against compression resistance	150 kN	267.56 kN	0.56	[1](Annex C)

7. Applicability

None.

8. Verification

Template struct.column-buckling@1.1.0 Content hash 9620a0aa904c2b4c Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. British Standards Institution. Structural Use of Steelwork in Building—Part 1: Code of Practice for Design—Rolled and Welded Sections. BS 5950-1:2000. BSI; 2000.